glTF as a World-Model Transport

Evidence-backed findings from an end-to-end ML pipeline

Khronos 3D Formats WG presentation

rudybear · github.com/rudybear/glTFWorldModel · MIT

Every finding in this deck has a code pointer and a measurement in the public repo.

Why this experiment

- World models & robotics need a **scene-state interchange**: simulation → training → inference → rendering
- Today's reality: everyone invents a container
  - Physion (NeurIPS benchmark): bespoke HDF5 + MP4 + CSV
  - Habitat/ReplicaCAD: **GLB for geometry + URDF sidecar** — because glTF lacks joints
  - OpenUSD: UsdPhysics 1.0 shipped in core
  - No published project uses glTF as an ML scene-state transport
- Question: **how far does glTF 2.0 + draft extensions get — what exactly is missing?**
- Method: **build the whole pipeline; record every impedance mismatch**

What we built (all open source, MIT)

```
MuJoCo sim ──▶ GLB episodes ──▶ headless renderer ──▶ rgb/seg/depth
(ground truth)  pose animation      (vendored pyrender)      │
                + KHR_physics_rigid_bodies (draft)           ▼
                + KHR_implicit_shapes (draft)      perception & dynamics
                + RWM_state_series (custom)          models (PyTorch)
                + semantics in extras                        │
                        ▲                                    ▼
                        └──── inference re-emits valid GLB ──┘
```

- 10 milestones, every one gated by **independent adversarial verification**
- 12,000+ generated episodes, 150 real Physion trials converted
- **Every emitted GLB passes the pinned Khronos glTF-Validator with 0 errors**
- Stock viewers (three.js, Babylon sandbox) play every episode — extensions ride along additively

What glTF got RIGHT for this use (5 positive findings)

1. **Accessor/bufferView machinery is a general typed time-series transport** — our custom state channels reuse it unchanged; zero schema failures across 10k+ episodes
2. **Additive extension model works**: `extensionsUsed` (never `Required`) keeps every file loadable by tools that know nothing about physics or state
3. **Single-file GLB episodes**: atomic, diffable, validator-checkable artifacts — GT and model predictions are directly comparable documents
4. **STEP animation sampling** honestly represents sampled simulator states
5. **Fixed units & coordinate conventions** (meters, seconds, Y-up RH) eliminated a whole class of ambiguity

Gap Part A — Core glTF has no concept of dynamic state

Gap	What's missing	Our workaround
G1	Velocity, action, uncertainty, joint state — **any** non-pose per-frame quantity	`RWM_state_series` (custom): named channels over ordinary accessors sharing the animation's time accessor
G2	Physics initial conditions (mass, friction, colliders)	draft KHR extensions (next slide)
G3	Channels wider than VEC4	documented chunking convention
G6	Uncertainty representation **and semantics**	diagonal-variance channel + a measured warning (slide 8)

Key point: ratifying the physics extensions does **not** close G1 — a time-series extension is a *separate, complementary* need.

Gap Part B — Implementing against the draft KHR physics extensions

We implemented `KHR_physics_rigid_bodies` + `KHR_implicit_shapes` (pinned draft commit) for real scenes, including articulated doors/drawers via limit-composed hinge/slider joints. Real gaps hit:

Gap	Finding	Cost we paid
G7	**No collider local offset/center** in KHR_implicit_shapes	forced a pivot-child-node design; unavoidable mesh-pivot vs collider-center error in real-asset conversion
G8	Limit damping is soft-stop only — **no viscous joint damping**	articulated scenes need side-channel metadata; MJCF/URDF have this natively

Gap Part B (continued)

Gap	Finding	Cost we paid
G9	Drives model persistent spring-to-target — **no bounded-duration push**	scripted actuation not encodable as a KHR drive at all
G10	**No fixed/weld joint**	handle attachment is derived, not constrained
G11	Single friction model vs static+dynamic pairs	measured information loss converting Physion (e.g. 1.0 vs 0.1 collapsed)

Gap Part C — Real-world conversion evidence (Physion)

Converted 150 trials of a real NeurIPS physics benchmark (ThreeDWorld HDF5) into the transport:

- 14 documented impedance mismatches: missing normals, mesh pivot conventions, camera matrix chirality, no ground-plane object concept, dropped collision events, friction collapse…
- **Result: 150/150 validator-clean GLBs; poses/velocities bit-exact round-trip**
- State-based label reconstruction: **92% agreement** with the benchmark's own labels — the transport carries enough state to reproduce a published benchmark's ground truth
- Honest negative: our small dynamics model does **not** transfer zero-shot (chance level) — the transport works; transfer learning is future work

A measured warning about uncertainty channels (G6)

Closed-loop experiment, 3 arms: oracle state / oracle + i.i.d. noise matched to measured perception error / real perception in the loop.

arm	pos error @ h=99
oracle state	0.36 m
oracle + **i.i.d.** noise	**27.6 m**
real perception loop	**1.62 m**

Real detector errors are **frame-correlated** (lag-1 autocorrelation 0.55–0.82) and largely cancel in finite-difference velocity estimation. An i.i.d.-calibrated uncertainty channel **overestimates closed-loop degradation 17×** — any future uncertainty extension should carry or at least warn about temporal correlation.

Recommendations (ranked by measured impact × blocking severity)

1. **Ratify a KHR time-series/state extension** alongside (not inside) the physics extensions — the `{node|joint|scene}`-target + shared-time-accessor + named-channel pattern is validated here at 10k-episode scale
2. **Collider local offset in KHR_implicit_shapes** — narrowest fix, outsized payoff
3. **Joint viscous damping + bounded drive mode** — parity with MJCF/URDF for ordinary actuation
4. **Fixed/weld joint type**

Recommendations (continued)

5. **Optional mesh/convex-hull collider** — the cost of its absence is measured in our real-asset conversion
6. **Best-practice guidance: per-frame uncertainty ≠ i.i.d. noise license** (measured 17× cost)
7. Second friction coefficient; root-level gravity

Explicitly *not* recommended: video-frame sequences in glTF; widening accessors past VEC4 — conventions suffice.

External validity: we tested ourselves

Our own verification protocol only proves we built what we said we built — so we ran two experiments designed to check the claims *from the outside*:

- **Blind spec-only reimplementation.** Zero source access — only `RWM_EXTENSIONS.md` + schemas + GLBs — decoded a whole episode **bitwise-identically**. It had to guess 6 conventions our docs left implicit; one was initially *wrong* (silently-wrong shapes). All 6 are now normative.
- **Clean-room reproduction from the public clone.** A fresh `git clone` + documented setup reproduced our smoke-test pass/skip counts and split sizes **digit-for-digit**, and seeded dataset generation **bit-identical** across machines.

This is exactly what a ratification process exists to surface — ambiguities an author blind to their own tacit assumptions can't see in their own writing. Two cheap experiments found 6 normative gaps in a spec we thought was complete. That's the argument for taking the

Everything is verifiable

- **Public repo**: github.com/rudybear/glTFWorldModel (MIT)
- `docs/GAP_REPORT.md` — all 20 gaps + 5 positives, each with code pointer, measurement, JSON exhibit, prior-art comparison (UsdPhysics / URDF / MJCF)
- `docs/VERIFICATION.md` — every claim re-runnable: exact commands + expected outputs
- Every milestone was gated by **independent adversarial verification** (the verifier famously caught our own overclaims — the corrections are visible in the docs, per project policy)
- Sample episodes: drag any generated GLB into a stock viewer — it plays

Ask: feedback on recommendation #1 (time-series extension scoping) and #2 (collider offset) — we'd contribute the RWM_state_series experience to a KHR-track effort.

Thank you

github.com/rudybear/glTFWorldModel

Questions — or better: clone it and run `docs/VERIFICATION.md` against us.